Dear Hiring Team,

I am applying for the Thermal CFD Simulation Engineer position at PACCAR. What draws me to this work is developing mathematical and physics-based models from first principles and understanding the physics behind their predictions. The position's focus on heat transfer, flow, and simulation-to-test correlation provides a concrete setting for applying that approach to thermal design decisions.

During my doctoral research at the University of Texas at San Antonio, I developed custom three-dimensional finite element solvers for structural mechanics, transient heat transfer, and fluid dynamics. This work included coupled structural-thermal habitat models and interior thermal-fluid simulations. Developing these models required translating the governing equations into numerical formulations and implementing their coupling. That foundation is relevant to evaluating how thermal and flow assumptions affect predicted system behavior, particularly when a model must support comparisons between design alternatives rather than provide a single numerical result.

At Rheem, I applied CFD to optimize fan-shroud geometry and improve airflow distribution across a condenser coil. I then fabricated and tested a prototype that demonstrated more than 80% improvement in airflow efficiency, measured as CFM/W, relative to the previous design. This project provides practical evidence of using simulation to guide a physical design and checking its performance experimentally. Although the application was HVAC rather than an exhaust system, the work connects directly to PACCAR's need to turn simulation findings into design recommendations supported by test data.

My implementation experience also includes a real-time C/C++ HVAC plant model developed at L&T Technology Services for John Deere, combining transient thermal and fluid formulations. I would bring this modeling and implementation background to your thermal analysis team while developing the exhaust-specific and tool-specific knowledge required by the role. I would welcome the opportunity to discuss how that experience could contribute to PACCAR's work.

Sincerely,
Adnan Shahriar